

II B.Tech II Semester			
SUBJECT NAME: SOFTWARE ENGINEERING			R23
Code: 23AIT02			Question Bank
UNIT-I			
PART-A		2 MARKS	
CO	BTL		
1.	Define software engineering.	CO1	L1
2.	State any two reasons for the emergence of software engineering.	CO1	L2
3.	What is meant by exploratory style of software development?	CO1	L1
4.	List any two major changes in modern software development practices.	CO1	L2
5.	What is a software development project?	CO1	L1
6.	Mention any two characteristics of a good software life cycle model.	CO1	L2
7.	What is the main objective of the Waterfall model?	CO1	L1
8.	Identify one limitation of the Waterfall model.	CO1	L4
9.	What is Rapid Application Development (RAD)?	CO1	L2
10.	How does Agile development handle changing requirements?	CO1	L4
PART-B		10 MARKS	
1.	Explain the evolution of software and discuss the factors that led to the emergence of software engineering as a discipline.	CO1	L4
2.	Describe the various types of software development projects. Explain the challenges involved in managing them.	CO1	L3
3.	Explain the exploratory style of software development. Compare it with traditional development approaches.	CO1	L4
4.	Discuss the notable changes in software development practices over the years with suitable examples.	CO1	L3
5.	Explain computer system engineering and its relationship with software engineering.	CO1	L2
6.	Describe the concept of software life cycle models. Explain the phases involved in a typical life cycle model.	CO1	L3
7.	Explain the Waterfall model in detail. Discuss its advantages, disadvantages, and applications.	CO1	L4
8.	Describe Rapid Application Development (RAD) model. Explain its process, advantages, and limitations.	CO1	L3
9.	Explain the Agile development model. Discuss Agile principles and benefits over traditional models.	CO1	L4
10.	Explain the Spiral model with a neat diagram. Analyze how risk management is handled in this model.	CO1	L3

UNIT-II				
PART-A		2 MARKS	CO	BTL
1.	Define software project management.		CO2	L1
2.	Mention any two complexities involved in software project management.		CO2	L2
3.	List any two responsibilities of a software project manager.		CO2	L1
4.	What is meant by project size estimation?		CO2	L2
5.	What is COCOMO?		CO2	L1
6.	What is Halstead’s Software Science?		CO2	L2
7.	Define risk management in software projects.		CO2	L3
8.	What is Software Requirements Specification (SRS)?		CO2	L1
9.	State one advantage of formal system specification.		CO2	L4
10.	Why is requirements analysis considered a critical phase in software development?		CO2	L3
	<u>PART-B</u>		10 MARKS	
1.	Explain software project management. Discuss the major complexities faced during software project execution.		CO2	L4
2.	Describe the roles and responsibilities of a software project manager in detail.		CO2	L2
3.	Explain various software project size estimation techniques used in project planning.		CO2	L4
4.	Discuss empirical estimation techniques used in software project estimation.		CO2	L3
5.	Explain the COCOMO model. Describe its types and applications in software cost estimation.		CO2	L4
6.	Explain Halstead’s Software Science. Discuss its importance in software measurement.		CO2	L3
7.	Describe software risk management. Explain the steps involved in identifying and controlling project risks.		CO2	L4
8.	Explain the process of requirements gathering and analysis with suitable examples.		CO2	L3
9.	Describe Software Requirements Specification (SRS). Explain its structure and characteristics.		CO2	L2
10.	Explain formal system specification techniques. Discuss axiomatic and algebraic specifications with examples.		CO2	L4

UNIT-III				
PART-A		2 MARKS	CO	BTL
1.	Define software design.		CO3	L1
2.	State any two characteristics of a good software design.		CO3	L2
3.	What is meant by layered arrangement of modules?		CO3	L1
4.	Define cohesion in software design.		CO3	L2
5.	What is coupling?		CO3	L1
6.	What is agility in software development?		CO3	L2
7.	Mention any two principles of Extreme Programming (XP).		CO3	L1
8.	What is structured analysis?		CO3	L2
9.	What is a Data Flow Diagram (DFD)?		CO3	L3
10.	Why is user interface design important in software systems?		CO3	L4
	<u>PART-B</u>		10 MARKS	
1.	Explain the software design process. Discuss the characteristics of a good software design.		CO3	L4
2.	Describe layered arrangement of modules. Explain cohesion and coupling and their significance in software design.		CO3	L3
3.	Explain different approaches to software design with suitable examples.		CO3	L4
4.	Discuss agility and the cost of change in software development. Explain how Agile processes address these issues.		CO3	L3
5.	Explain the Agile process model. Describe the key practices involved in Agile development.		CO3	L2
6.	Explain Extreme Programming (XP). Discuss its values, practices, advantages, and limitations.		CO3	L4
7	Describe Function-Oriented Software Design using SA/SD methodology.		CO3	L3
8.	Explain structured analysis. Develop a DFD model for a simple system and explain its components.		CO3	L4
9.	Explain structured design and detailed design. Discuss the importance of design review.		CO3	L3
10.	Explain user interface design. Discuss characteristics of a good UI and various types of user interfaces.		CO3	L4

UNIT-IV			
PART-A		2 MARKS	
CO	BTL		
1.	Define coding in software engineering.	CO4	L1
2.	What is code review?	CO4	L2
3.	State any two objectives of software documentation.	CO4	L1
4.	Define software testing.	CO4	L3
5.	What is black-box testing?	CO4	L2
6.	What is white-box testing?	CO4	L3
7.	What is debugging?	CO4	L1
8.	What is integration testing?	CO4	L2
9.	What is smoke testing?	CO4	L3
10.	Why is software quality management important in software projects?	CO4	L4
PART-B		10 MARKS	
1.	Explain the importance of coding standards and principles in software development.	CO4	L4
2.	Describe the code review process. Discuss its benefits and limitations.	CO4	L3
3.	Explain software documentation. Describe different types of documentation used in software projects.	CO4	L2
4.	Explain software testing. Discuss the objectives and principles of testing.	CO4	L4
5.	Differentiate between black-box testing and white-box testing with examples.	CO4	L3
6.	Explain debugging techniques and program analysis tools used in software testing.	CO4	L4
7.	Describe integration testing. Explain various integration testing strategies.	CO4	L3
8.	Explain testing of object-oriented programs. Discuss the challenges involved.	CO4	L4
9.	Explain software reliability. Discuss statistical testing techniques used to measure reliability.	CO4	L3
10.	Explain software quality and software quality management system (SQMS). Discuss their role in delivering quality software.	CO4	L4

UNIT-V				
PART-A		2 MARKS	CO	BTL
1.	Define CASE in software engineering.		CO5	L1
2.	What is the scope of CASE tools?		CO5	L2
3.	What is a CASE environment?		CO5	L1
4.	List any two characteristics of CASE tools.		CO5	L2
5.	What is meant by second-generation CASE tools?		CO5	L3
6.	Define software maintenance.		CO5	L1
7.	What is software reverse engineering?		CO5	L2
8.	Mention any two characteristics of software maintenance.		CO5	L3
9.	What is software reuse?		CO5	L1
10.	Why is software reuse not widely practiced in organizations?		CO5	L4
	<u>PART-B</u>		10 MARKS	
1.	Explain CASE and its scope. Discuss the role of CASE tools in software development.		CO5	L4
2.	Describe the CASE environment. Explain the architecture of a CASE environment with neat diagram.		CO5	L3
3.	Explain CASE support in various phases of the software life cycle.		CO5	L4
4.	Discuss the characteristics of CASE tools. Explain the need for second-generation CASE tools.		CO5	L3
5.	Explain software maintenance. Discuss its types and characteristics in detail.		CO5	L4
6.	Explain software reverse engineering. Discuss its objectives and benefits.		CO5	L3
7.	Describe various software maintenance process models. Explain how maintenance cost is estimated.		CO5	L4
8.	Explain software reuse. Discuss the reasons for limited reuse in software projects.		CO5	L3
9.	Explain different approaches to software reuse with suitable examples.		CO5	L4
10.	Discuss software reuse at organizational level. Explain the issues involved in implementing a reuse program.		CO5	L3